

EMAIL THREAT FORENSIC INVESTIGATION REPORT

OFFICIAL DIGITAL FORENSIC EXAMINATION DOSSIER // LAW ENFORCEMENT & SOC RECORD

Case Identifier	Examination Date	Investigation Status	Evidence Hash (SHA-256)
CASE-002	2026-09-06	VERIFIED EXAMINATION	9e1c6a3f8b2d0475c0e9a4f1b7d3...
Classification: BEC	Risk Score: 84 / 100	Sample: FW_Urgent_Vendor_...	File Size: 48 KB

1.0 EXECUTIVE SUMMARY

A forensic examination was conducted on evidence sample 'FW_Urgent_Vendor_Account_Change.eml' cataloged under reference CASE-002. The electronic message claims transmission from sender 'Ananya Mehta' (ananya.mehta@meridian-infra.com). Automated and deterministic protocol evaluation assessed an authoritative risk score of 84 out of 100, resulting in a forensic threat classification of BEC. Authentication protocol verification recorded SPF: PASS, DKIM: PASS, and DMARC: PASS. No direct lookalike domain registration was detected in the immediate IOC telemetry. Routing telemetry analyzed 3 relay hops from origin to terminal destination server.

2.0 EMAIL & EVIDENCE SPECIFICATIONS

Evidence Field	Forensic Data Value	Verification Finding
Original Filename	FW_Urgent_Vendor_Account_Change.eml	Binary structure parsed and pres...
File Size	48 KB	Standard envelope size
Cryptographic Hash	9e1c6a3f8b2d0475c0e9a4f1b7d3c6e8a2f5b0d4c7e1a9f...	SHA-256 cryptographic integrity ...
Claimed Sender Name	Ananya Mehta	Header-extracted display name
Claimed Sender Email	ananya.mehta@meridian-infra.com	Header From address
Terminal Mail Server	Bengaluru, India	Terminal destination host
Originating Node	197.210.53.88	First recorded external IP hop

3.0 AUTHENTICATION PROTOCOL ANALYSIS

Standard email authentication protocols were checked against published Domain Name System (DNS) records. Discrepancies indicate that the sending infrastructure was not authorized to transmit on behalf of the domain.

Protocol	Verdict	Technical Examination Finding
SPF (Sender Policy Framework)	PASS	Sending host verified against authorized SPF record
DKIM (DomainKeys Identified)	PASS	Valid cryptographic signature verified against DNS public key
DMARC (Domain Alignment)	PASS	Message satisfies alignment policy for domain

4.0 HEADER & ROUTING INFRASTRUCTURE

The message envelope traversed 3 recorded network transit hops. Network infrastructure was cross-referenced with autonomous system records and datacenter threat feeds.

Hop	IP Address	Geographic Location	Relay Timestamp	Forensic Infrastructure Note
#1	197.210.53.88	Lagos, Nigeria	2026-09-06 06:45:10	Session origin before relay
#2	40.107.8.52	Singapore	2026-09-06 06:45:32	Compromised Microsoft 365 tenant r...
#3	49.207.50.12	Bengaluru, India	2026-09-06 06:45:49	Destination mail server

5.0 LANGUAGE, INTENT & CONTENT ANALYSIS

Content linguistics and social engineering vectors were scored at 30 out of 30 maximum allocated points. Extracted textual phrases were analyzed for deception patterns, urgency cues, and financial solicitation.

Category / Signal	Detected Phrasing / Pattern	Confidence Assessment
FINANCIAL DIVERSION	"please update payment beneficiary immediately"	94.0% Match
URGENCY	"process the attached wire transfer before end of day"	89.0% Match

6.0 REPUTATION & DOMAIN INTELLIGENCE

Attribute	Observed Evidence	Analyst Interpretation
Domain Age	4018 days	Domain age within normal operational thresholds
Typosquatting Target	None observed	No typosquatting target observed
Reputation Score Contribution	14 / 15 pts	Elevated risk: Associated infrastructure flagg...

7.0 INDICATORS OF COMPROMISE (IOCs)

The following actionable technical indicators were extracted during examination for perimeter blocklisting and SIEM correlation:

Indicator Type	Technical Value	Context / Threat Finding
IP Address (Origin)	197.210.53.88	Originating relay host observed in he...
Email Address	ananya.mehta@meridian-infra.com	Header From address
Forensic Artifact	IFSC:HDFC0002891	Extracted payment/URL artifact
Forensic Artifact	IBAN:DE89370400440532013000	Extracted payment/URL artifact
File Hash (SHA-256)	9e1c6a3f8b2d0475c0e9a4f1b7d3c6e8a2f5b0d4c7e1a...	Unique cryptographic hash of message ...

8.0 ATTRIBUTION ASSESSMENT

ATTRIBUTION FINDING: LIKELY COMPROMISED LEGITIMATE ACCOUNT

Authentication protocols validated successfully; however, anomalous content indicates account takeover.
Note: Attribution conclusions represent technical forensic assessments based on available telemetry and do not constitute definitive legal identification without formal judicial subpoena records.

9.0 CORRELATION & INFRASTRUCTURE RELATIONSHIPS

Graph analysis cross-referenced entities across historical telemetry clusters and previous incident dossiers:

Source Entity	Target Entity	Correlation Basis / Linkage
email1	ip1	same sending IP
email2	email1	thread hijack
domain1	email1	compromised mailbox

10.0 RISK ASSESSMENT & SIGNAL WEIGHTING

Overall Risk Score: 84 / 100 (Threshold evaluation: CRITICAL RISK). The score is derived across four standardized examination quadrants:

Quadrant Vector	Allocated Weight	Observed Score	Analytical Contribution
Header & Routing	30 pts	22 pts	Evaluates relay anomaly, ASN hosting, and h...
Authentication Failure	25 pts	18 pts	Evaluates SPF, DKIM signature validity, and...
Language & Intent	30 pts	30 pts	Evaluates urgency, financial diversion, and...
Reputation & History	15 pts	14 pts	Evaluates domain age, typosquatting targets...

11.0 KEY INVESTIGATION FINDINGS

- 1. High-confidence threat signature: Overall threat score exceeds enterprise containment threshold.
- 2. Authentication verified: Cryptographic signatures and domain alignment passed checks.
- 3. Domain registration integrity: No deceptive lookalike typography detected.
- 4. Origin routing: Outbound transmission originated from standard commercial/workspace relay.

12.0 RECOMMENDED INVESTIGATION ACTIONS

- 1. Evidence Preservation: Retain full raw RFC 5322 message headers and message body in forensic storage.
- 2. Network Containment: Implement perimeter firewall block on originating IP 197.210.53.88.
- 4. Incident Response: Query mail gateway logs for all inbound messages matching the sending domain within past 30 days.
- 5. Account Safeguards: Enforce credential reset and session termination for any recipient who engaged with message content.

13.0 AUTOMATED NARRATIVE GENERATION ARCHITECTURE NOTE

SYSTEM NOTICE: This document was produced via deterministic evidence extraction from validated email forensic JSON. The platform architecture is designed to integrate text-generation language models in future backend iterations to draft extended contextual DFIR narratives while maintaining the underlying JSON as the single source of truth.

14.0 EVIDENCE ATTESTATION & LEGAL DISCLAIMER

REPORT IDENTIFIER: REP-CASE-002 | **PLATFORM:** Email Threat Intelligence & Forensic Platform | **EVIDENCE SOURCE:** Ingested MIME/MSG Artifact | **STATUS:** VERIFIED FORENSIC RECORD

LEGAL DISCLAIMER: This document is an analytical examination record compiled from electronic message headers, cryptographic authentication tokens, DNS telemetry, and network routing hops. Findings should be corroborated with original raw headers, provider server logs, and judicial legal process before presentation in legal proceedings.